

Nexivarin 10 mg Tablets — eCTD Submission
Module 2 — CTD Summaries

Submission Type	NDA — 505(b)(1)	Sequence Number	0000
Submission Date	2021-03-15	Section	2.2
Prepared by	PharmaCorp Inc.	Confidentiality	Proprietary and Confidential

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PharmaCorp Inc.
NDA 123456

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Section 2.2
CTD Introduction

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2.2 Introduction to the CTD Submission

2.2.1 Drug Product Overview

This NDA (NDA 123456) seeks approval of Nexivarin 10 mg Tablets for the treatment of hypertension and stable angina pectoris. Nexivarin hydrochloride is a novel dihydropyridine calcium channel blocker characterized by high bioavailability, a long elimination half-life enabling once-daily dosing, and minimal first-pass metabolism. The drug substance general information, nomenclature, and physicochemical properties are provided in **Section 3.2.S.1** of Module 3.

Refer to Section 3.2.S.1 (Module 3) for the complete nomenclature, structural formula, and general physicochemical properties of nexivarin hydrochloride, including BCS classification (Class II) and solubility data.

2.2.2 Drug Development History

The development history of nexivarin is summarized in Table 2.2-1 below. Detailed nonclinical pharmacology and toxicology studies referenced herein are reported in full in Module 4 (Sections 4.2.1–4.2.3). Clinical study reports supporting efficacy and safety are provided in Module 5 (Sections 5.3.3 and 5.3.4).

Year	Activity	Key Outcome
2011	Discovery of nexivarin scaffold	Patent filing US10,123,456
2013	Preclinical efficacy proof-of-concept	Antihypertensive activity confirmed in SHR model (see Section 4.2.1.1)
2014	IND Filing (IND 67890)	Phase I authorization granted
2015	Phase I SAD/MAD studies	Well-tolerated; PK established
2016	Phase I PK studies	Bioavailability 78%; t½ 38 hr (refer to Section 5.3.1)
2017	Phase II dose-ranging (NXV-201)	10 mg optimal dose identified (see Section 5.3.3.1)
2018–2020	Phase III pivotal studies (NXV-301, NXV-302)	Significant BP reduction; NI margin met (see Sections 5.3.3.2, 5.3.3.3)
2020	End-of-Phase 3 meeting with FDA	Agreement on submission package
2021-Q1	NDA Submission	This submission (Sequence 0000)

Table 2.2-1 above summarizes key milestones. Full nonclinical study reports are provided in Module 4 (Sections 4.2.1–4.2.3). Clinical study reports are filed in Module 5, Sections 5.3.1 through 5.3.4.

2.2.3 Regulatory Strategy

The NDA is submitted under 21 CFR §505(b)(1) as a fully originator application. The application relies exclusively on studies conducted by or for the Applicant. All documents are provided in PDF/A-1b format per FDA eCTD technical specifications (Version 3.2.2).

2.2.4 Overview of Clinical Development Program

Phase I: Single ascending dose (SAD) and multiple ascending dose (MAD) studies in 96 healthy volunteers established the PK profile of nexivarin. The complete bioavailability and food-effect study (NXV-BA-01) is provided in **Section 5.3.1**. Population PK analyses are summarized in **Section 2.7.2** (Clinical Pharmacology Summary).

As described in Section 5.3.1, absolute bioavailability was confirmed at 78.2% (CV 12%). No clinically significant food effect was observed (AUC ratio 1.04; refer to Table 1 in Section 5.3.1 for the food-effect data).

Phase II: A double-blind, randomized, placebo-controlled, dose-ranging study (NXV-201, Section 5.3.3.1) in 320 patients with mild-to-moderate hypertension evaluated doses of 2.5 mg, 5 mg, and 10 mg. The 10 mg dose demonstrated the optimal benefit/risk profile as depicted in Figure 2.2-1 (dose-response curve) presented in Section 2.7.3.

Phase III: Two pivotal, multicenter, double-blind, randomized, placebo-controlled studies (NXV-301 and NXV-302) each enrolling approximately 900 patients confirmed the efficacy of nexivarin 10 mg. Primary efficacy results are detailed in **Section 5.3.4.2**; the integrated safety data are provided in **Section 5.3.4**.

The Phase III efficacy and safety results are summarized in Section [2.5](#) of this Module 2 submission. Full clinical study reports are filed in Module 5, Sections [5.3.3.2](#) (NXV-301) and [5.3.3.3](#) (NXV-302). As shown in Table 1 of Section 5.3.4.2, nexivarin 10 mg achieved an LS mean SeDBP reduction of -8.6 mmHg vs. placebo ($p < 0.0001$).